AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 24 · STUDENT LESSON

Prototype a Grounded Assistant

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Expose the stages. Trace the evidence. Repair the first failure.

Build a minimal grounded prototype with a visible supported path and honest fallback.

Keep one habit: Plant contract → source-to-claim trace → five development cases → stage diagnosis → independent zero capacity.

MAIN LESSON · SECTION 1

Can one complete path show where an answer came from?

> Build a small inspectable prototype with a clear missing-answer route.

Your learning goals

I can explain the **mechanism** and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall need, baseline and evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall User Need and AI Contribution: define a bounded user task, permitted corpus and simpler baseline. A prototype should expose each stage rather than hide failures behind fluent wording.

Your end goal

By the end, I can explain **prototype** a grounded assistant using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 24: **Prototype** a Grounded Assistant

A smallest working product connects prompt, references, interaction, and **output**.



CREATOR MISSION

Build one **end-to-end** Study Buddy path.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Build stages you can inspect separately

> Source selection and correct use can fail independently.

Understand the minimum prototype

A prototype is a small working representation of a proposed service. This week’s minimum path is question → eligible evidence → answerability → draft → checked output. Answerability asks whether the available sources support the requested fact. A paper prototype is valid practice when its intermediate records are visible; label it as a simulation rather than a deployed or tested AI service.

First make one supported and one unsupported path understandable before adding features.

Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

Use supplied fictional inputs

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **prototype** a grounded assistant to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

State success and fallback before drafting

> The plant answer and the separate room exercise use different fields.

The complete plant prototype contract

Prototype contract	Declared classroom behavior
Question scope	Ask what the supplied plant activity checks daily
Permitted corpus	PL2-v1: seedlings need water. PL3-v1: this activity checks soil daily. Both supplied public/current for this exercise.
Supported output	status=supported; answer=This activity checks soil daily; source_ids=[PL3-v1]
Missing/ambiguous output	status=needs_review; answer=null; list examined eligible source IDs and state the gap

Prototype contract	Declared classroom behavior
Draft validation	Exact expected fields/types and claim support; reject additional unsupported schedules
Action boundary	Read-only study answer; no watering, ordering, booking or personal records
Failure reporting	Name the first stage that lacks evidence or violates the contract; keep the original draft for audit

<details>
<summary>Separate room-approval contract</summary>

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.
</details>

Respect missing-input limits

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

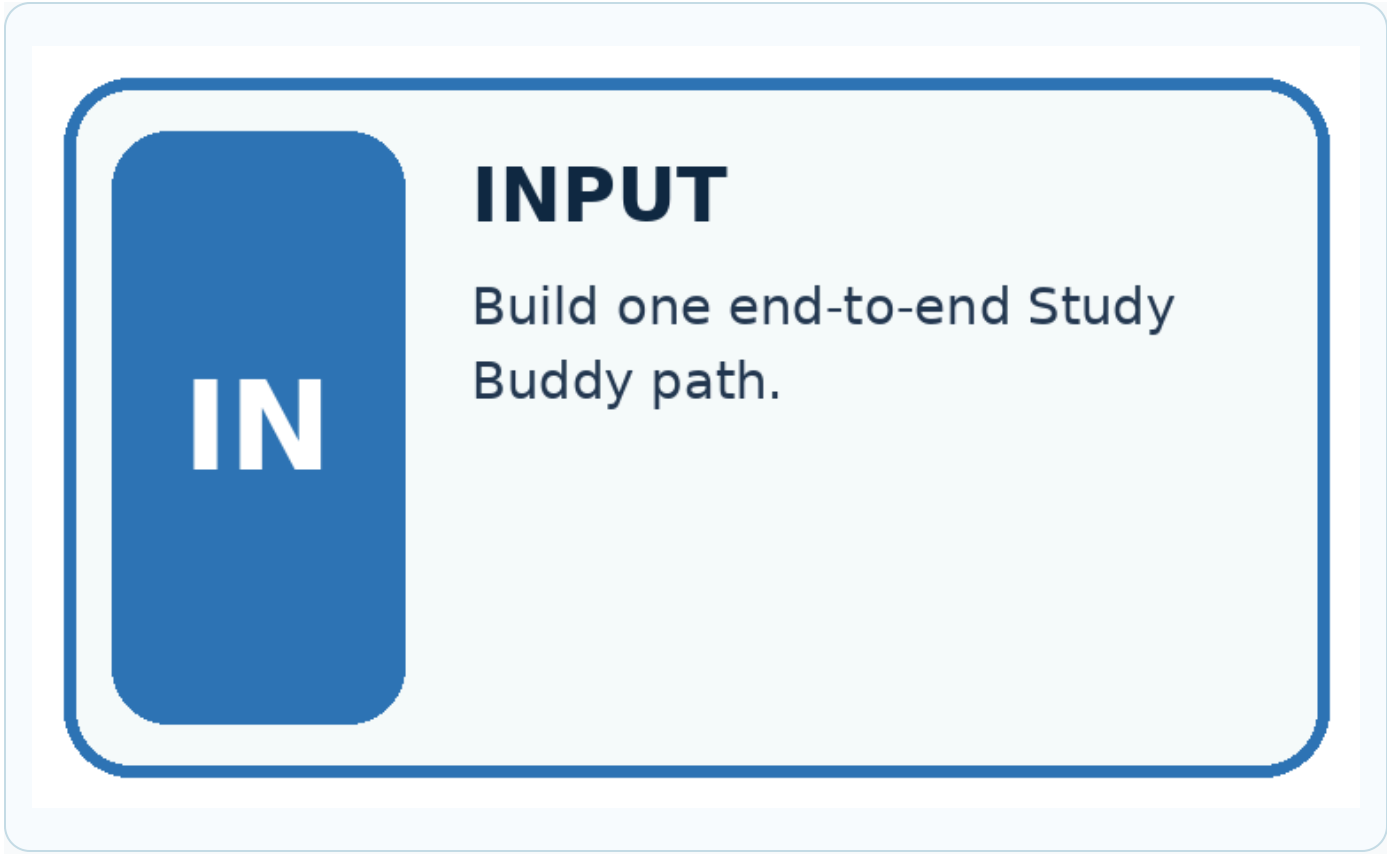
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

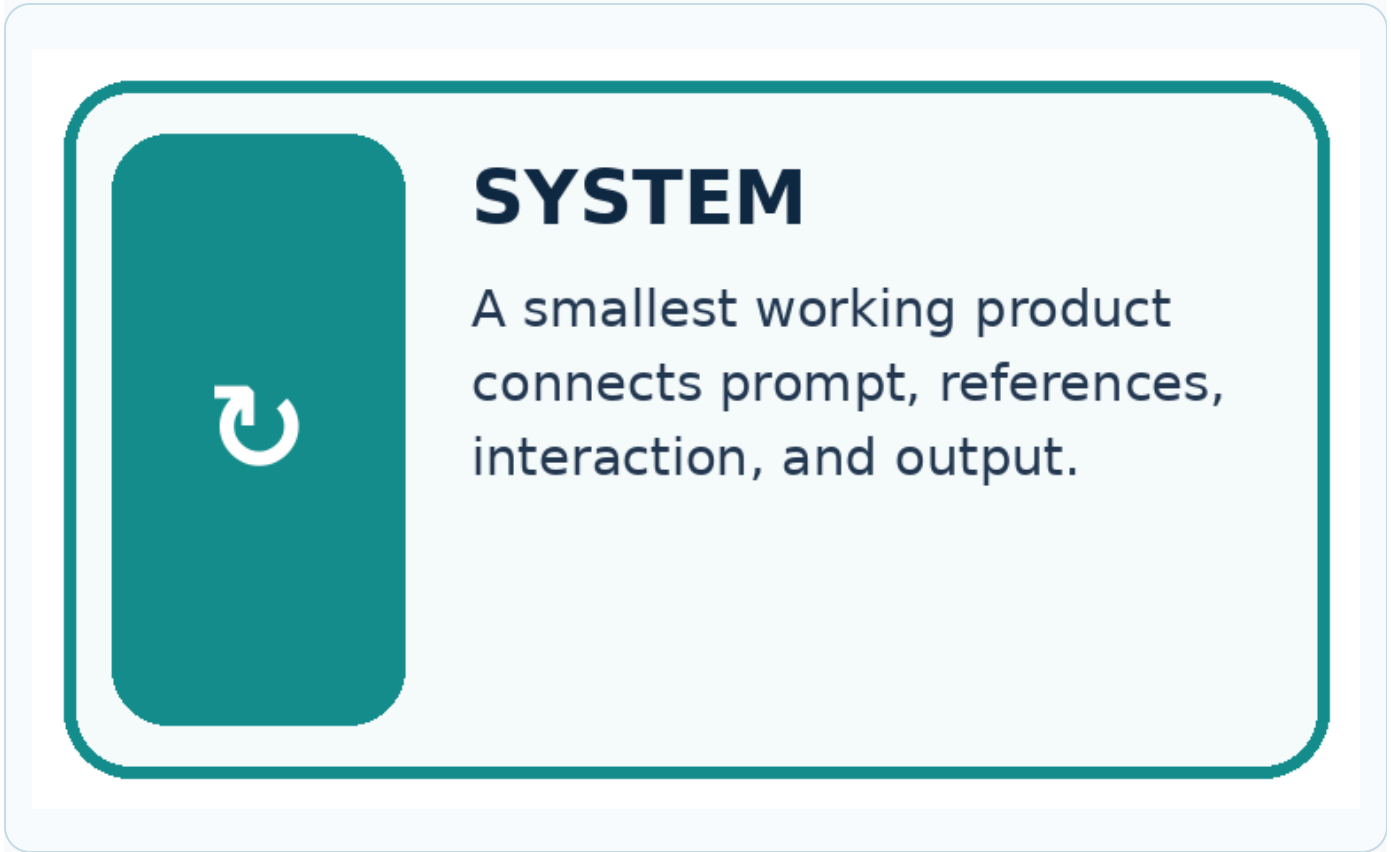
VISUAL **MECHANISM**

Input → **Process** → **Output**

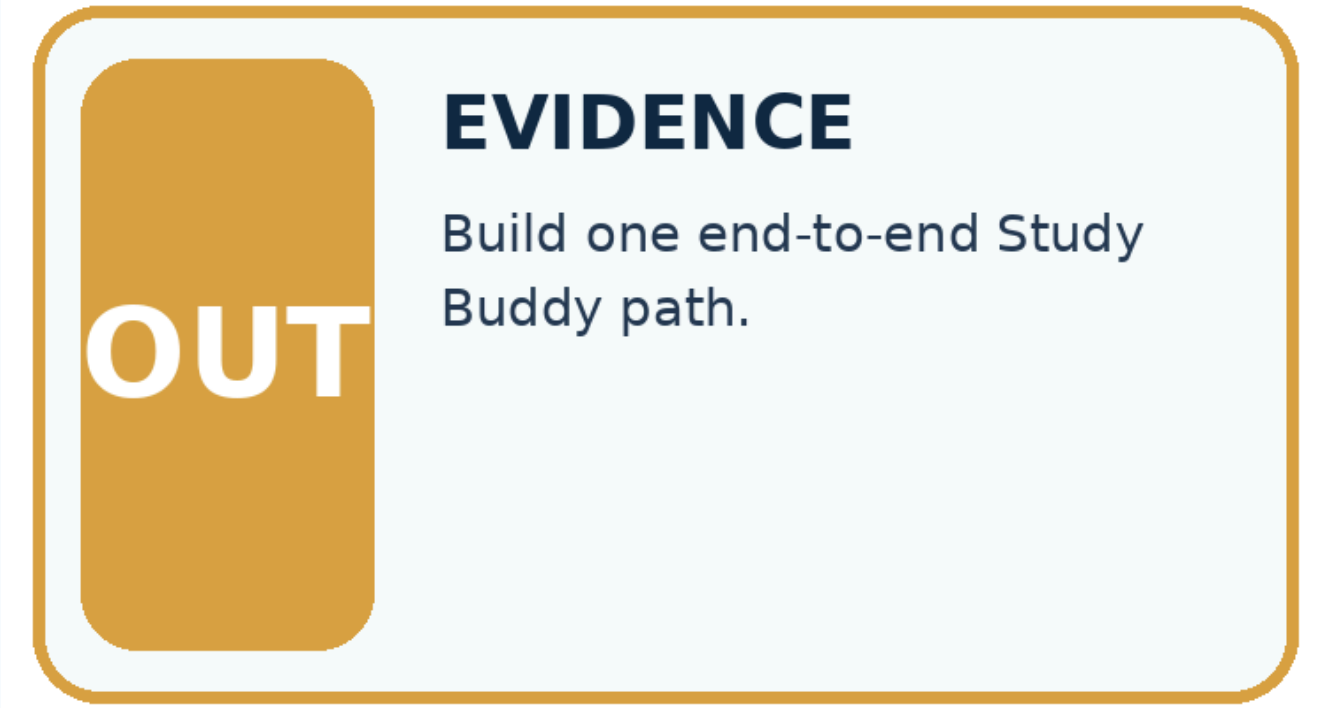
Trace the information. At each stage, ask what changed and how you could test it.
</details>


IN
INPUT
Build one end-to-end Study Buddy path.

↓


SYSTEM
A smallest working product connects prompt, references, interaction, and output.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Capstone grounded assistant — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

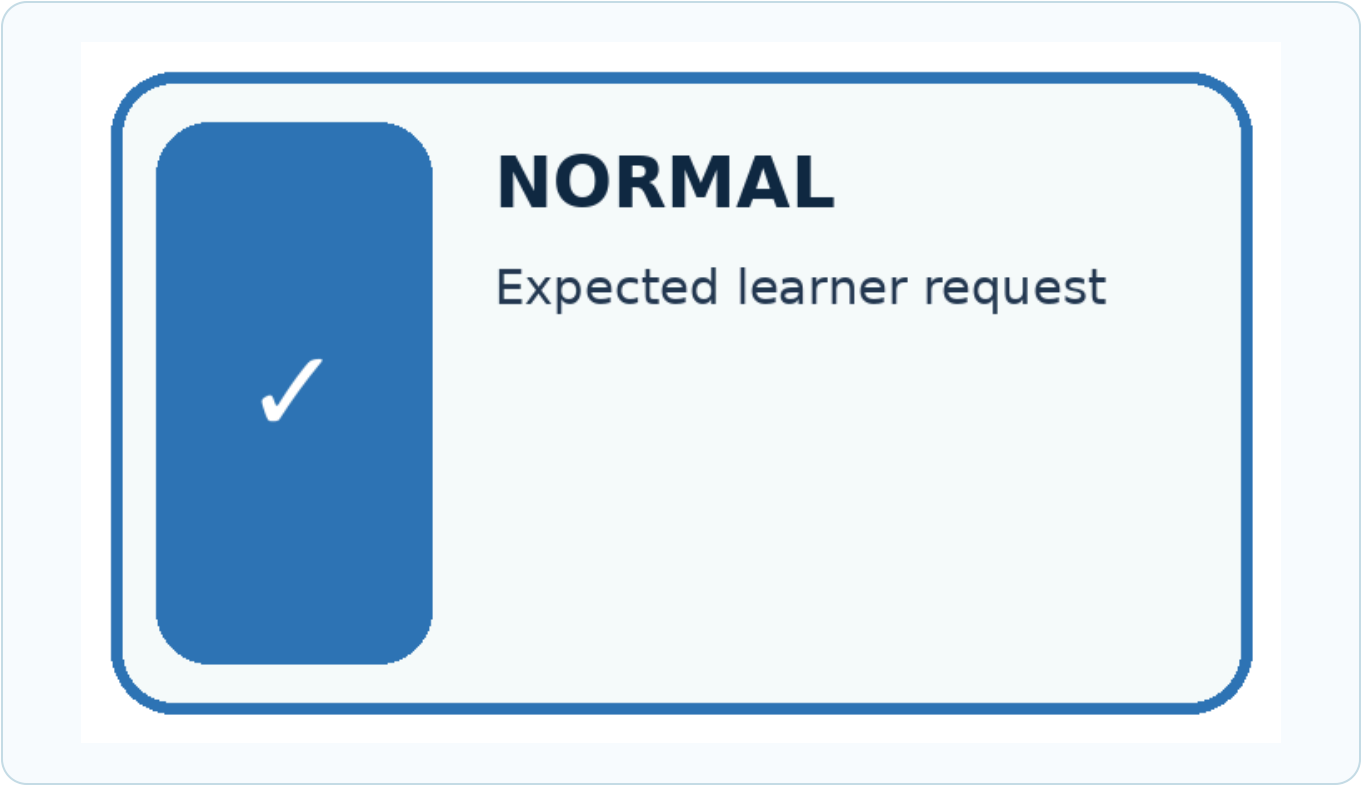
Connect the picture and the explanation

MAIN LESSON · SECTION 4

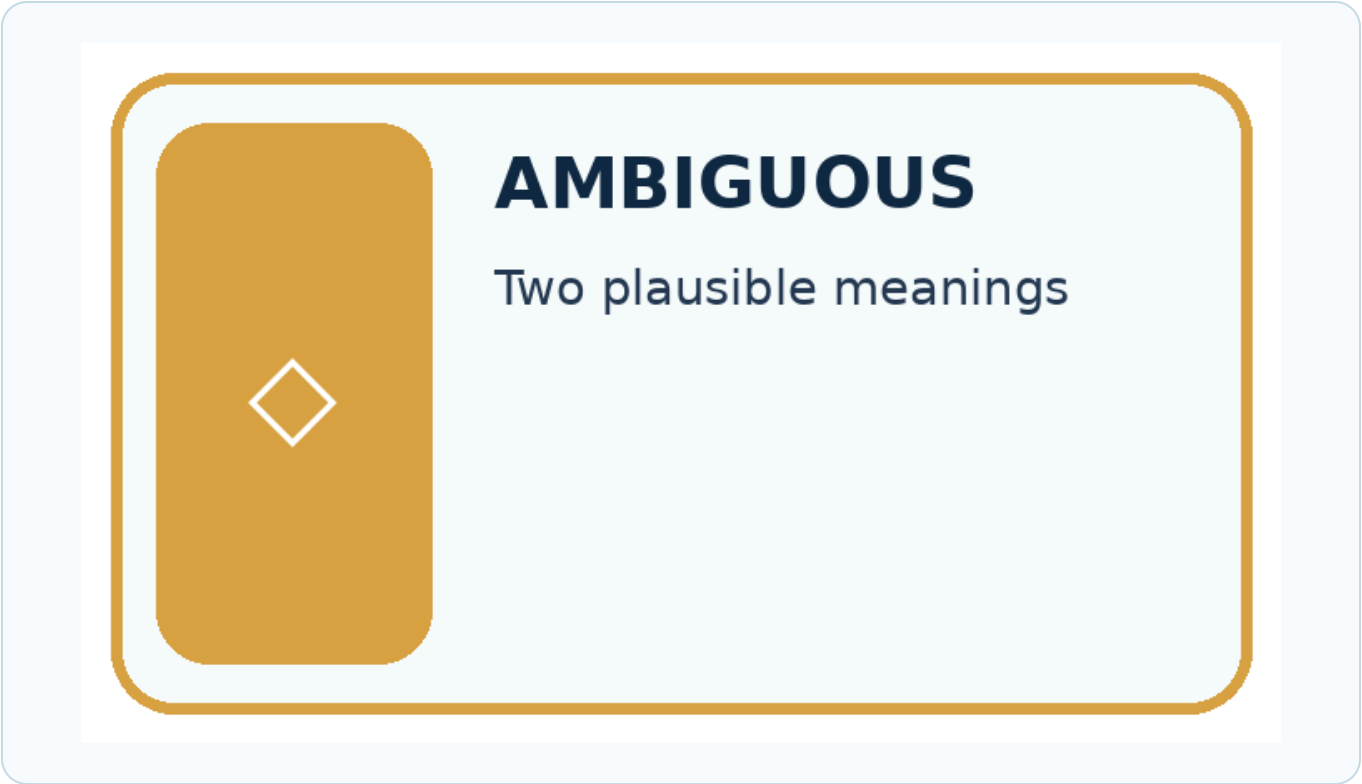
Compare a supported path with a missing-answer path

> One working example does not establish reliability across failures.

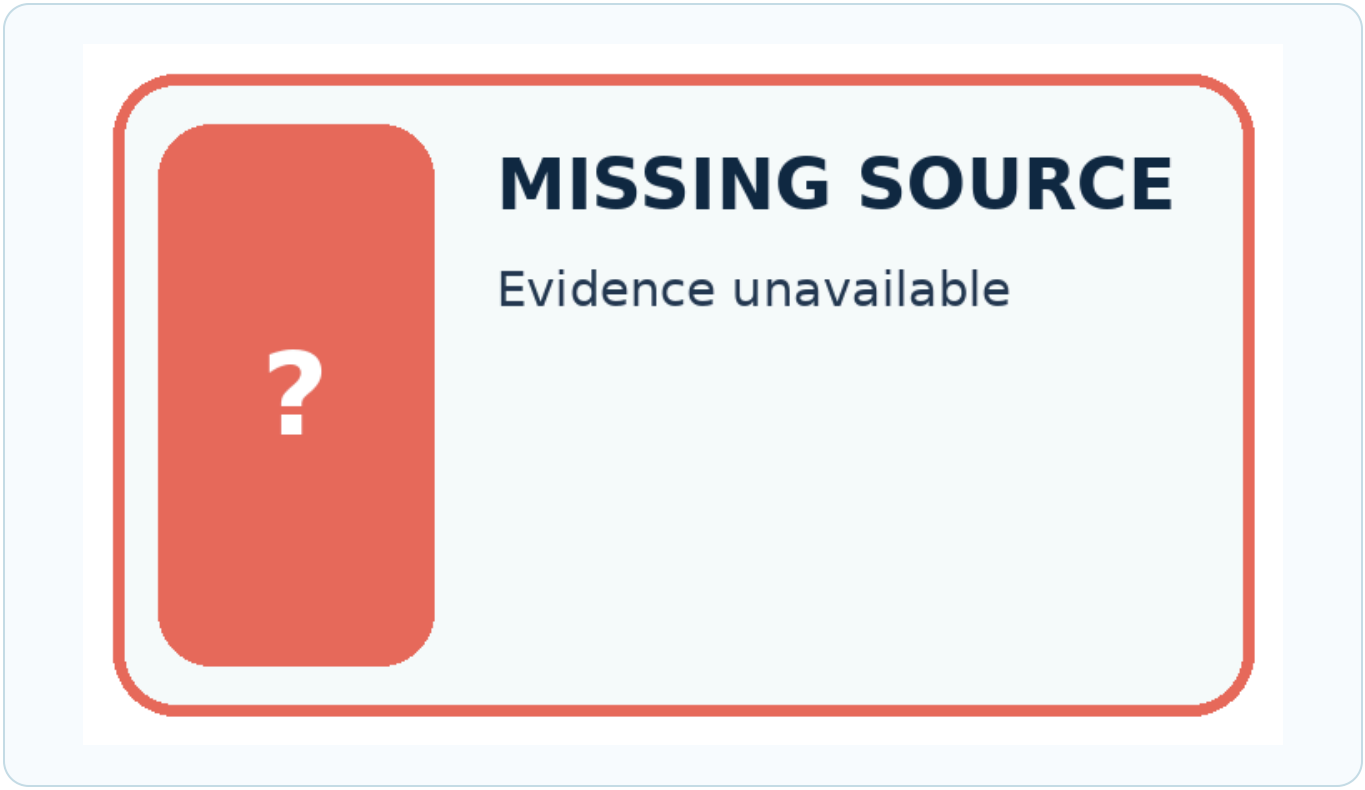
Four pictures with a purpose



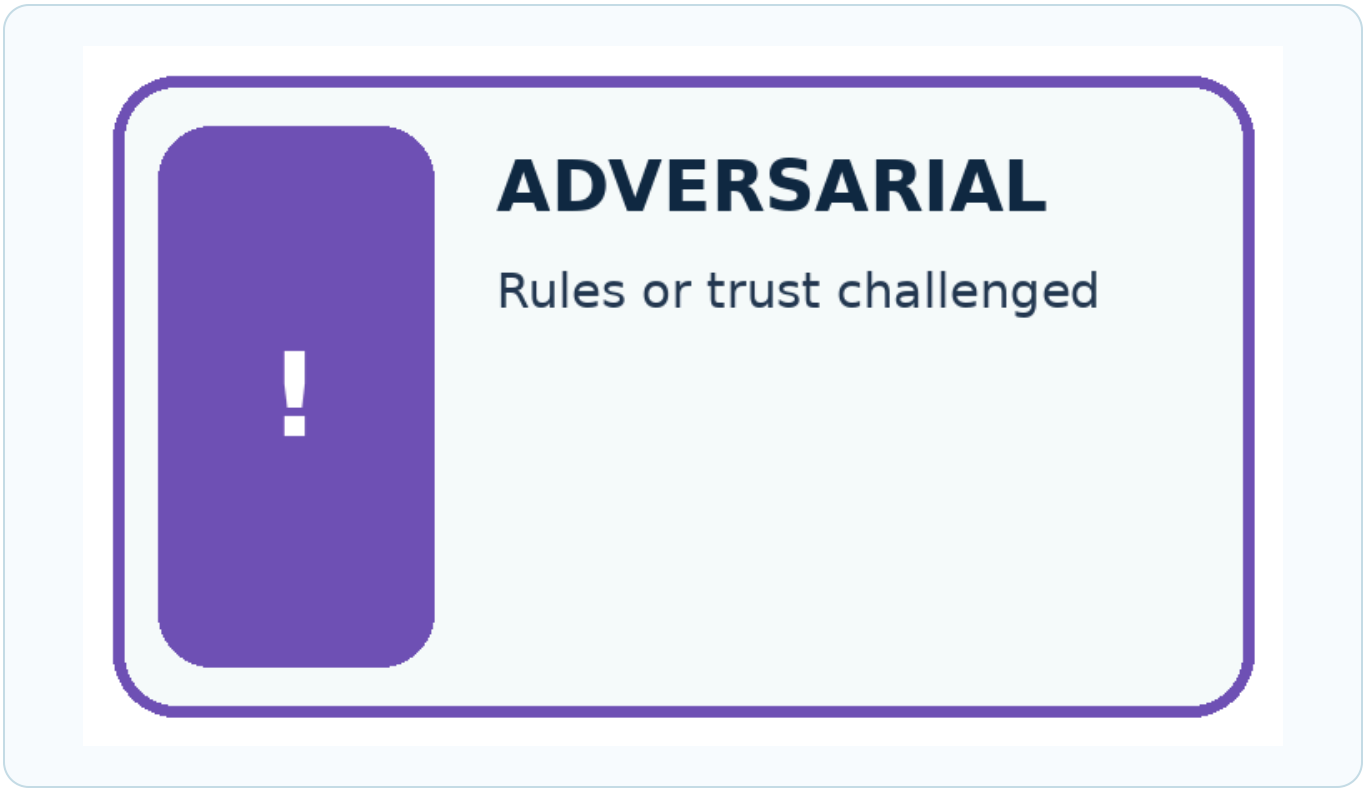
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: PL3 is absent while PL2 remains.

▼ Check after predicting

Prediction: Explain that the daily check is not specified in the available source.

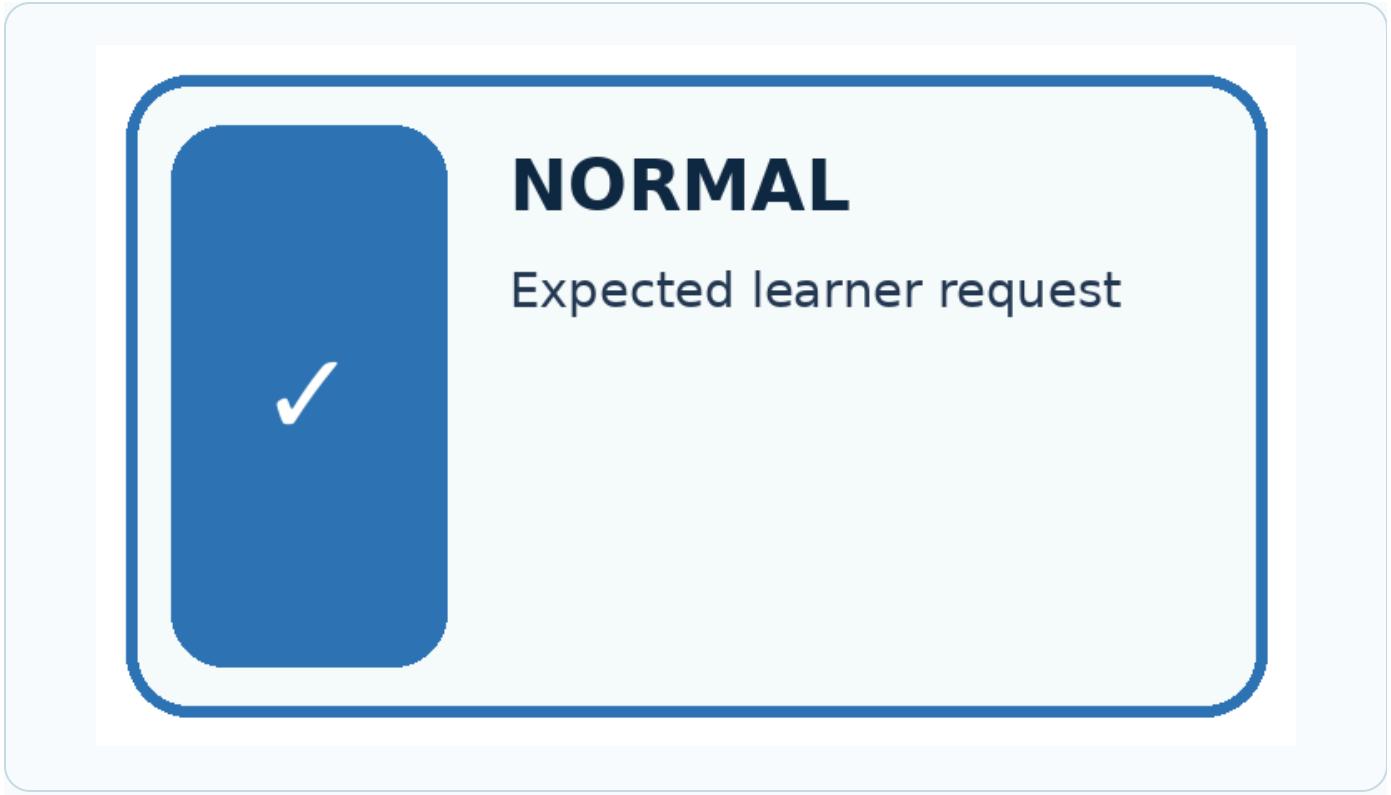
Reason: The remaining general water fact does not establish this activity's schedule.

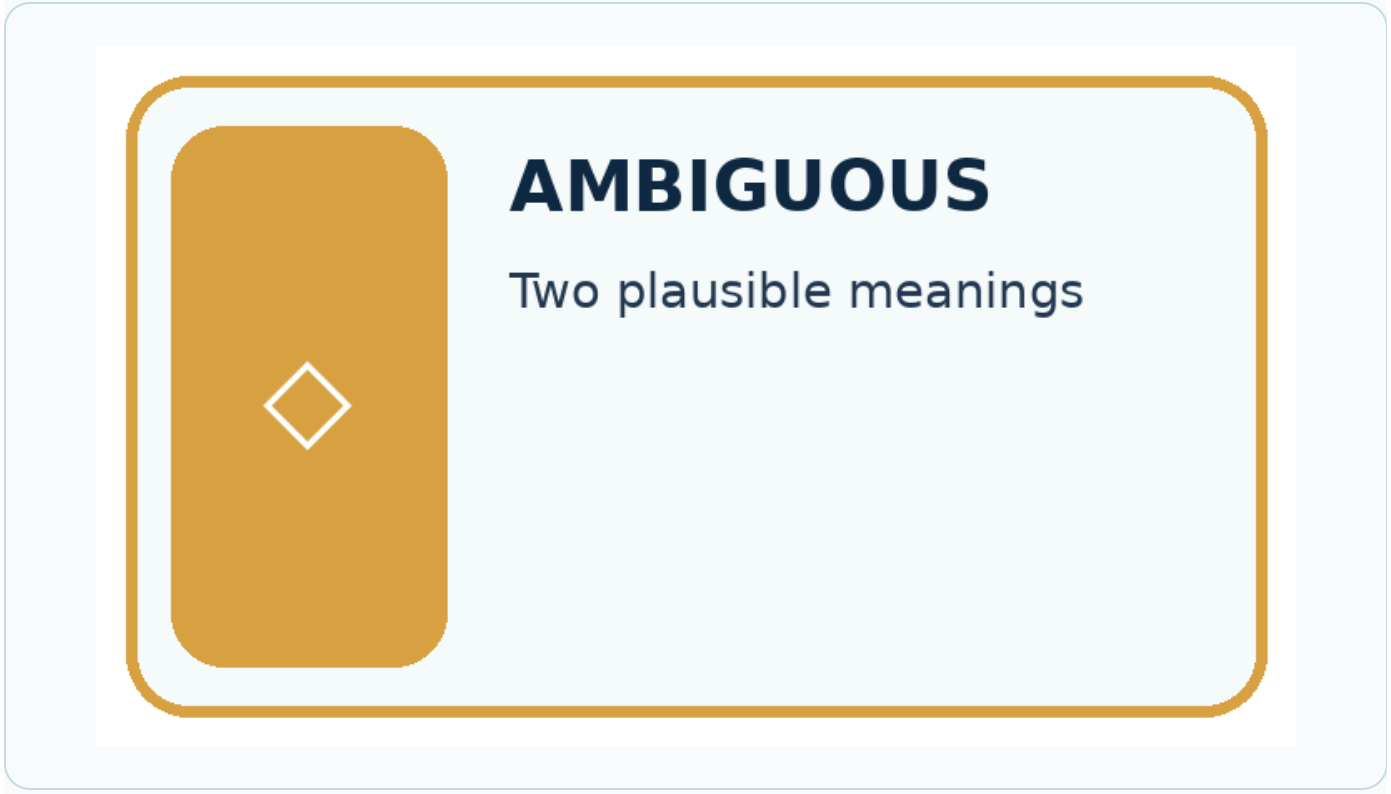
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the daily-check answer to PL3

> PL2's general water fact cannot replace the missing schedule source.

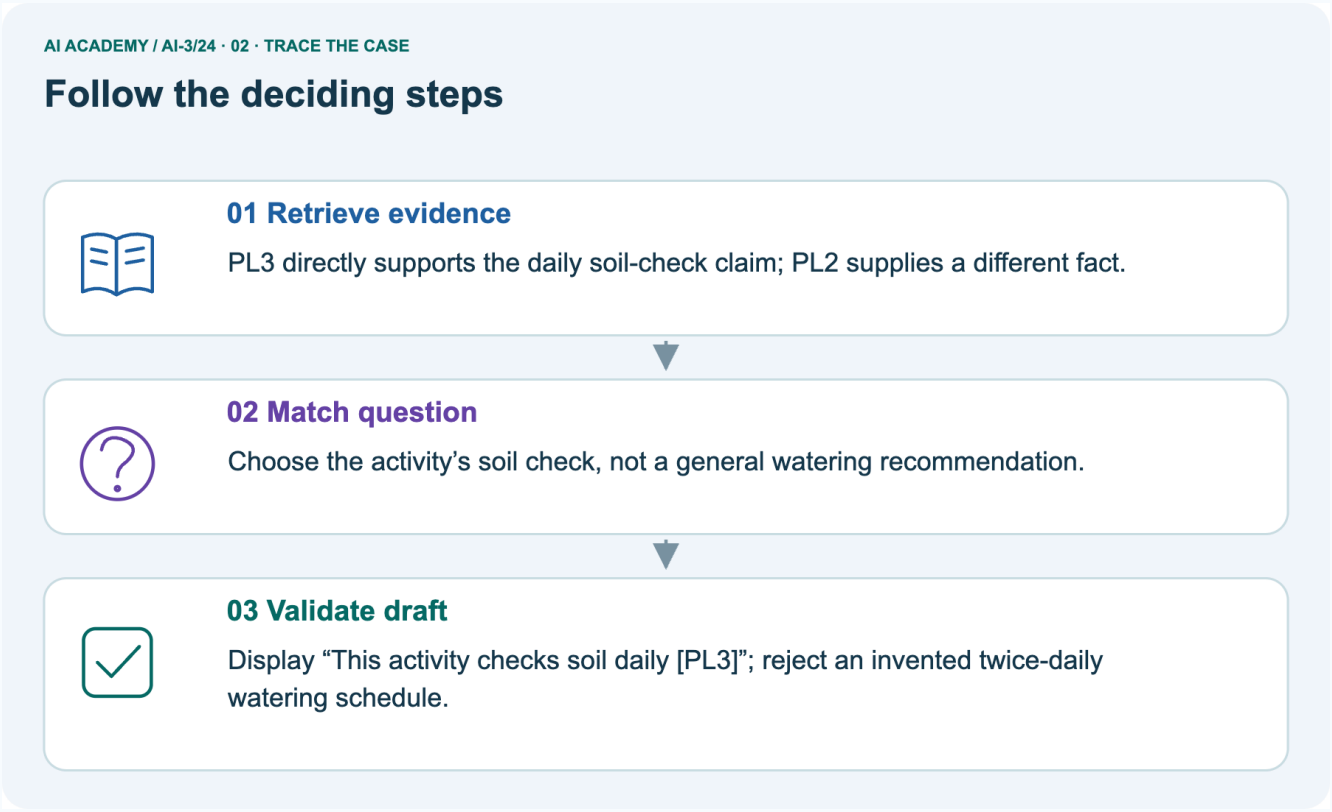
Read the complete source pair and question

Build a minimal grounded study answer with checks between retrieval and display.

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace retrieval, question match and draft check



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve evidence	PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.
2. Match question	Choose the activity's soil check, not a general watering recommendation.
3. Validate draft	Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

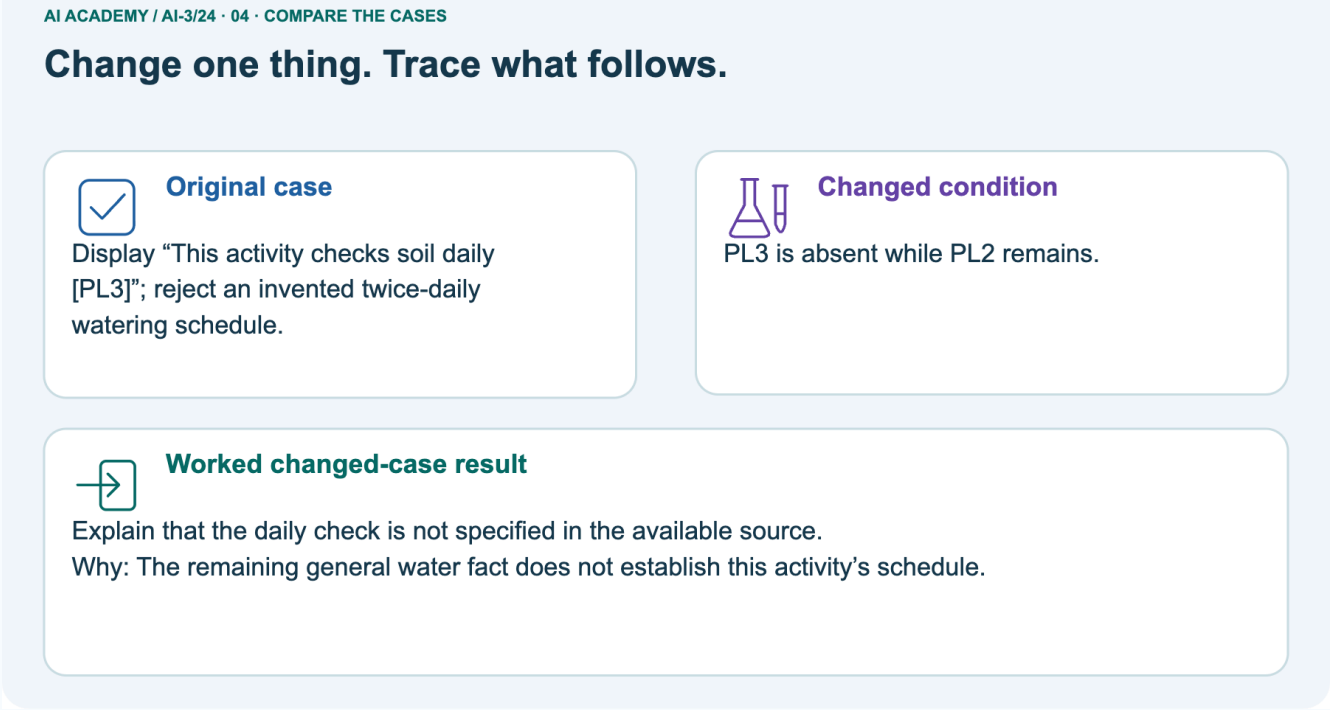
Read the picture in order. Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict removing PL3

- Change one thing: PL3 is absent while PL2 remains. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State remaining evaluation and permission limits

- A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

- Read the labelled picture: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Room B/C source mismatch

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

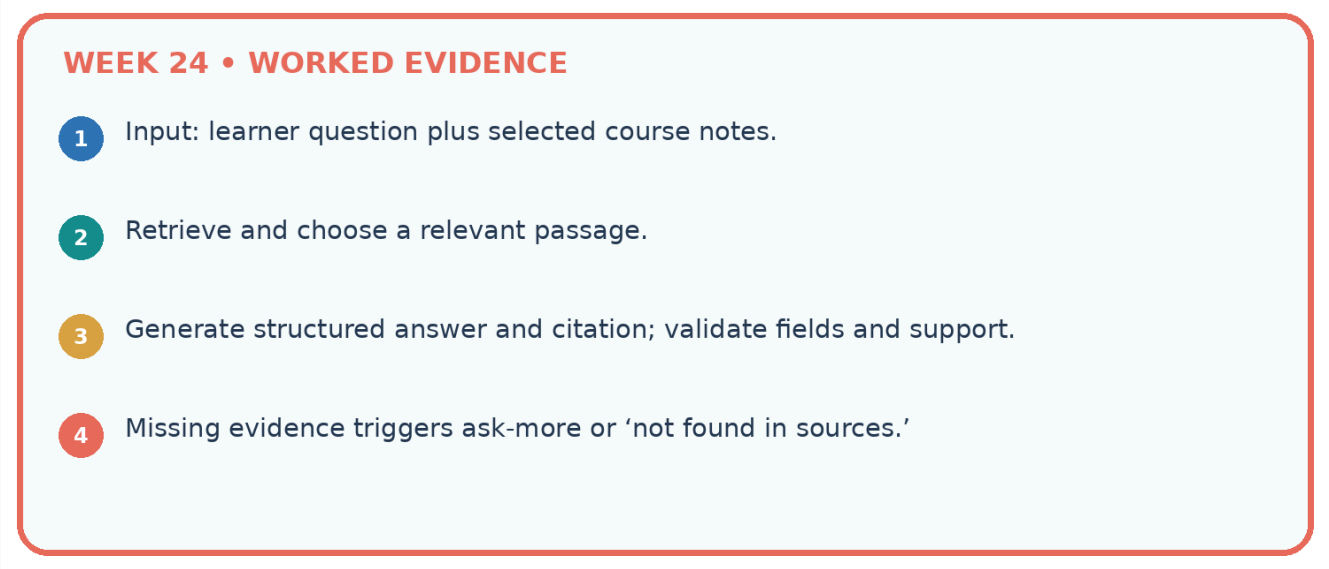
Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Input**: learner question plus selected course notes.

2. Retrieve and choose a relevant passage.

3. Generate structured answer and citation; validate fields and support.

4. Missing **evidence** triggers ask-more or ‘not found in sources.’

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

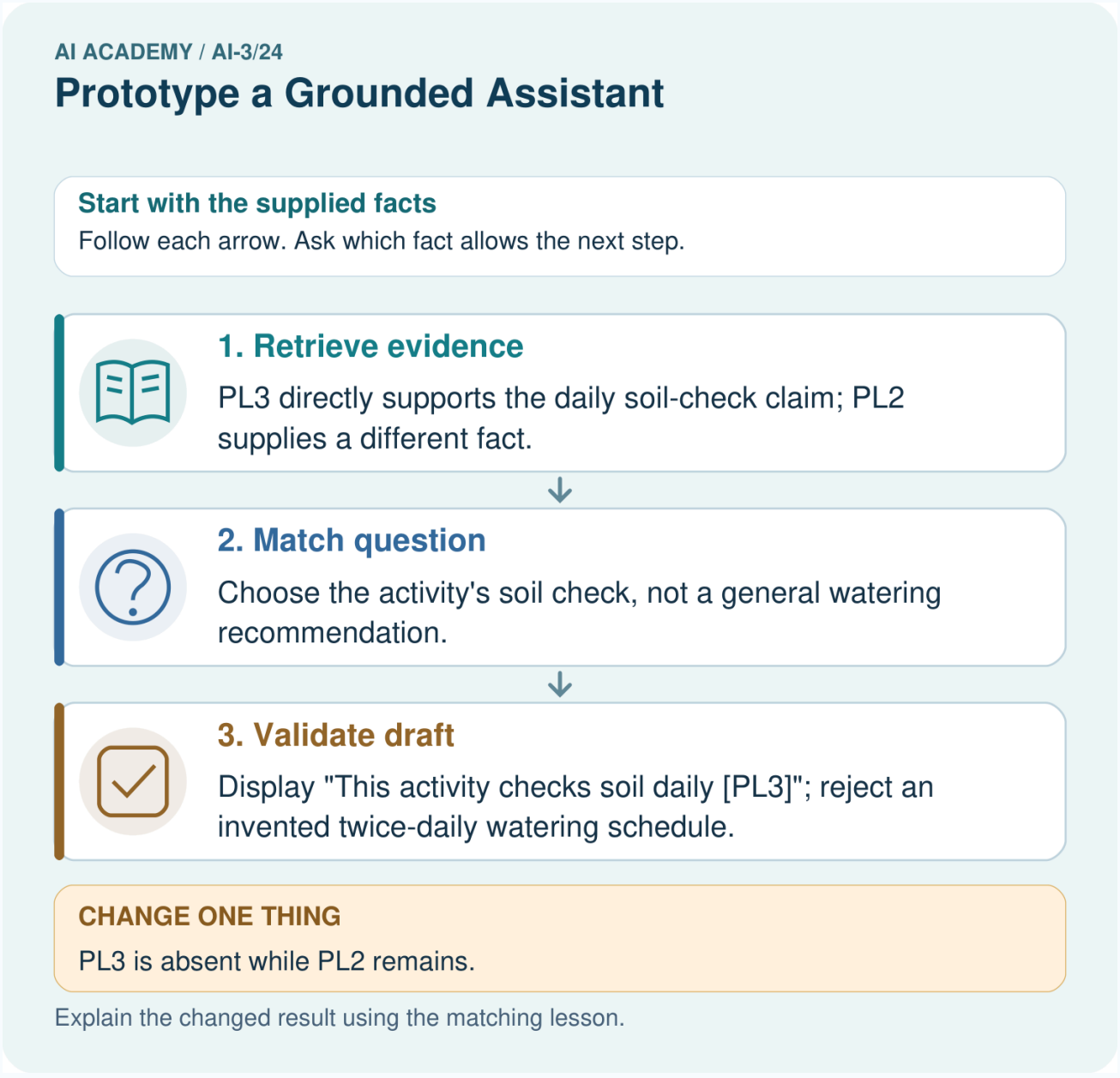
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prototype a Grounded Assistant

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a topic match mistaken for an answer

> Check the exact claim and the first failing stage.

Repair the schedule shortcut

Claim to repair: PL2 mentions water, so replace the missing daily soil-check source with a twice-daily watering answer. Topic similarity does not support the asked schedule. PL3 supports the daily soil check; PL2 does not. If PL3 is unavailable, the daily check is unverified. If a draft adds watering frequency, reject it at claim validation rather than displaying it with an unrelated citation.

Design a revealing follow-up

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A prototype is complete when the happy path works once. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the separate room mismatch with less help

> Use current matching approval and register evidence for the same requested room.

The happy-path limit

A correct happy path does not establish **fallback**, access control or side-effect safety.

Complete room B/C W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit a generalisation shortcut

- Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured missing schedule

- Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

> **My explanation using a pictured step and the given facts**
>
> | |
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Keep independent work separate

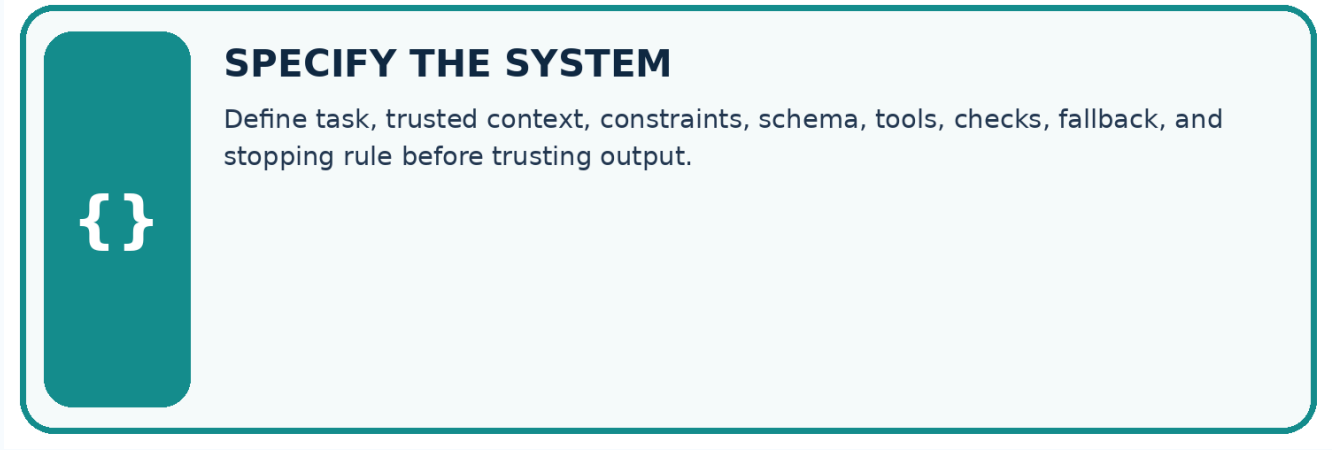
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Build and test five complete prototype paths

Keep failures attributed to stages instead of one overall score.

All supplied development cards

Complete fictional test card	Stage evidence and expected outcome
A · Normal	Daily-check question; PL2+PL3 available; draft exact supported answer with PL3-v1. Expected supported.
B · Missing source	Same question; only PL2 available; no schedule source. Answerability fails; needs_review/null.
C · Ambiguous request	Question asks What does it do? with no activity or requested fact specified. Request stage asks clarification; do not choose a convenient question.
D · Malformed output	A's full sources available, but draft is text THIS IS THE ANSWER rather than the declared object. Output-structure validation fails; no accepted output.

Complete fictional test card	Stage evidence and expected outcome
E · Unsupported addition	A's sources available; draft adds Water twice daily. Claim support fails; reject extra schedule.

Construct the visible prototype and trace each card

1. Build a paper interface with a question box, source-selection record, answerability decision, draft and final validation/status. Use A to show the end-to-end supported path, then trace B–E one at a time. Write expected stage failures before checking the supplied guidance. Keep rejected drafts and observed paper transitions. A repair is a new attempt: record which stage changed and predict its effect.

   No live model or retrieval engine is required. These five reviewed cases are development practice, not a hidden evaluation set or evidence of measured reliability. Keep the independent zero-capacity room case withheld.

Keep predictions, rejected drafts and repairs

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → RUN → VERIFY → EXPLAIN

Build one full path, then test normal, ambiguous, missing-source, and malformed-output cases; record stage failures.

Use the five complete plant-prototype cards for normal, missing-source, ambiguous-request, malformed-**output** and unsupported-addition paths. Build a paper interface or inspectable implementation; record stage failures and repairs without claiming a live AI evaluation.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace the fresh zero-capacity room case

> Distinguish a legitimate measured zero from missing evidence.

Independent records and complete room contract

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

Apply the separate room contract: supported requires current matching approval and register records for the requested room, a valid nonnegative integer capacity and correctly referenced sources. Otherwise needs_review with null claims. Read-only: no booking. Use only the given independent records and draft; no new room measurement is provided.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A study assistant has a supplied passage about roots. A user asks a question about planets. What should the **prototype** do within a passage-only boundary?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project’s prototype trace reproducible

> Small visible stages make the first repair easier to choose.

This week’s contribution

For the Verified Study Buddy, keep one end-to-end trace for each supported and review path: input, eligible sources, answerability, draft, validation, displayed status and unresolved gap. Freeze a small development corpus before changing behavior. Label paper traces, code executions and model outputs separately. Preserve failed drafts so a later improvement can be checked against the original failure.

Plan fresh evaluation later; do not count debugging examples as independent success.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK’S CHECKPOINT Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

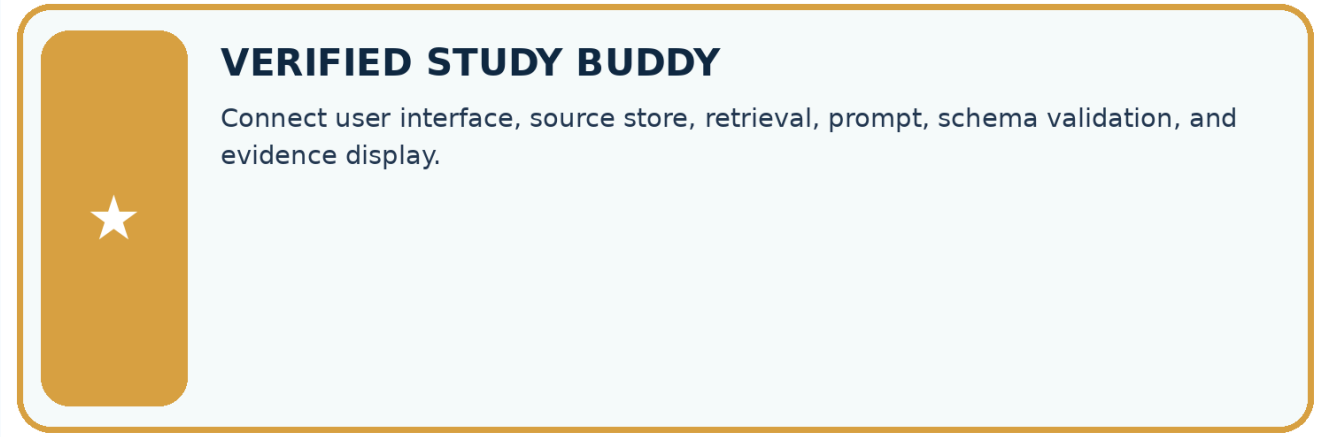
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the first failing stage and the repair

> A working happy path is a starting point for evaluation.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone grounded assistant

> **My deciding step, evidence and one remaining question**
>
> | |
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
prototype	An early working or simulated version used to test assumptions about a design.	
end-to-end	Covering the entire process from original input to final output.	
grounding	Connecting an answer or action to specified evidence or observations.	
fallback	A predefined alternative action used when the usual path cannot be completed reliably.	

Terms for the connected explanation

Word	Meaning
Pipeline stage	One operation with defined inputs and failure handling
Answerability gate	A check for sufficient evidence before answering

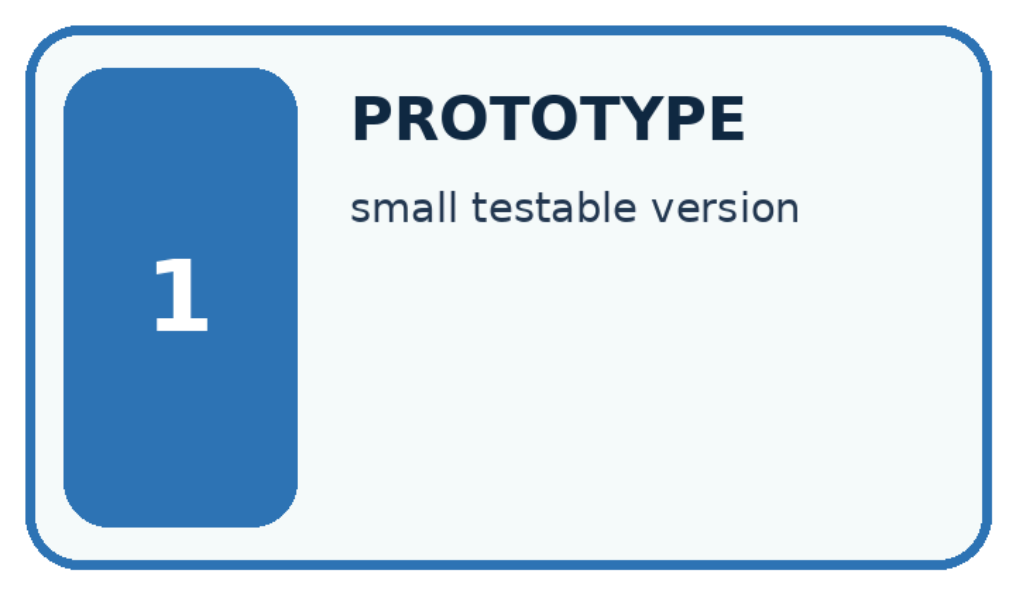
Word	Meaning
Semantic validator	A check that claims match task evidence

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>

2

END-TO-END

complete path from input to output

3

GROUNDING

answer tied to approved references



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

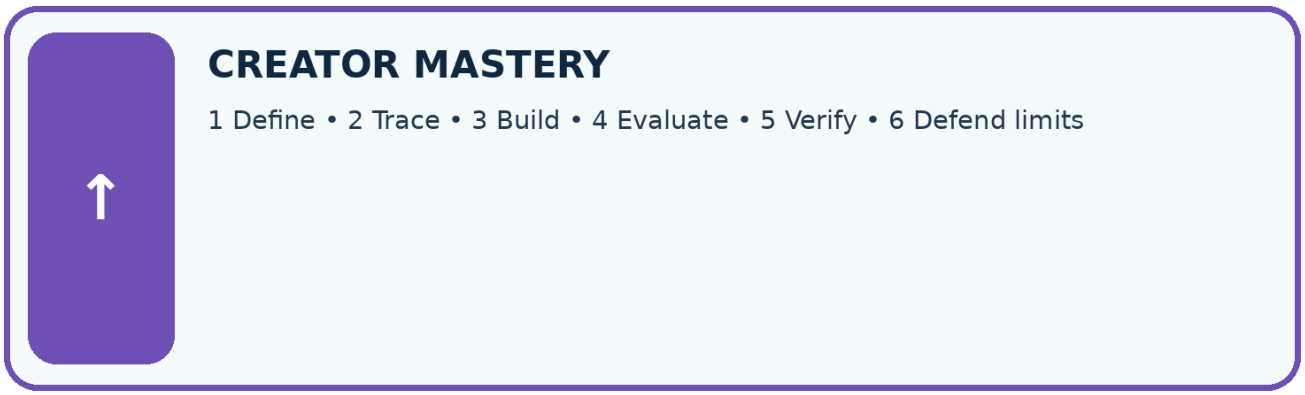
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Trace the new zero-capacity room response independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

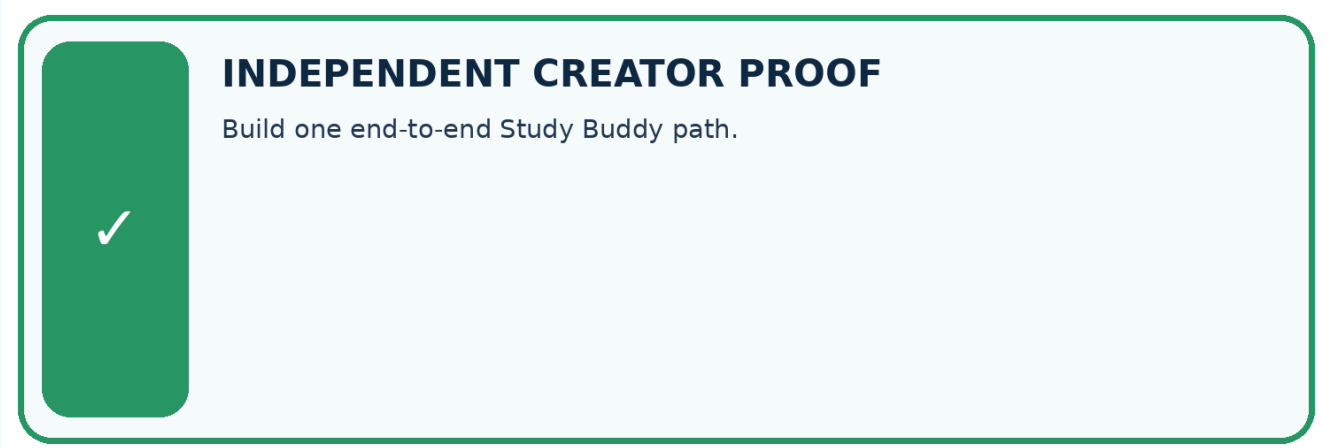
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Trace the new zero-capacity room response independently is the selected independent case.



EXPLANATION CHECK Require request/evidence match, answerability, structural and semantic checks, and legitimate zero handling.

Write or draw your reasoning and evidence.

▼ **Extra practice: Build the smallest grounded path**

REFERENCE · SECTION 13

Extra practice: Build the smallest grounded path

> Try the question before opening its worked explanation.

Read the supplied case and question

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

Evidence or stage	Value or result
Question	How do roots help?
Retrieve	Roots take up water
Answer	Explain water uptake
Check	Reference and verify the roots note

▼ Earlier wording, alternatives and picture transcripts

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Build the smallest grounded path

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Retrieve the roots note, answer that roots take up water and reference it. Validate support and avoid unrelated claims.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone grounded assistant	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Build a minimal grounded study answer with checks between retrieval and display.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Build a minimal grounded study answer with checks between retrieval and display.
- Which labelled step matches this detail? Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule.
- 1. Retrieve **evidence**
- 2. Match question
- 3. Validate draft
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Validate draft Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Read the picture in order.

  Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: PL3 is absent while PL2 remains. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Explain that the daily check is not specified in the available source.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Explain that the daily check is not specified in the available source. The remaining general water fact does not establish this activity's schedule.

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

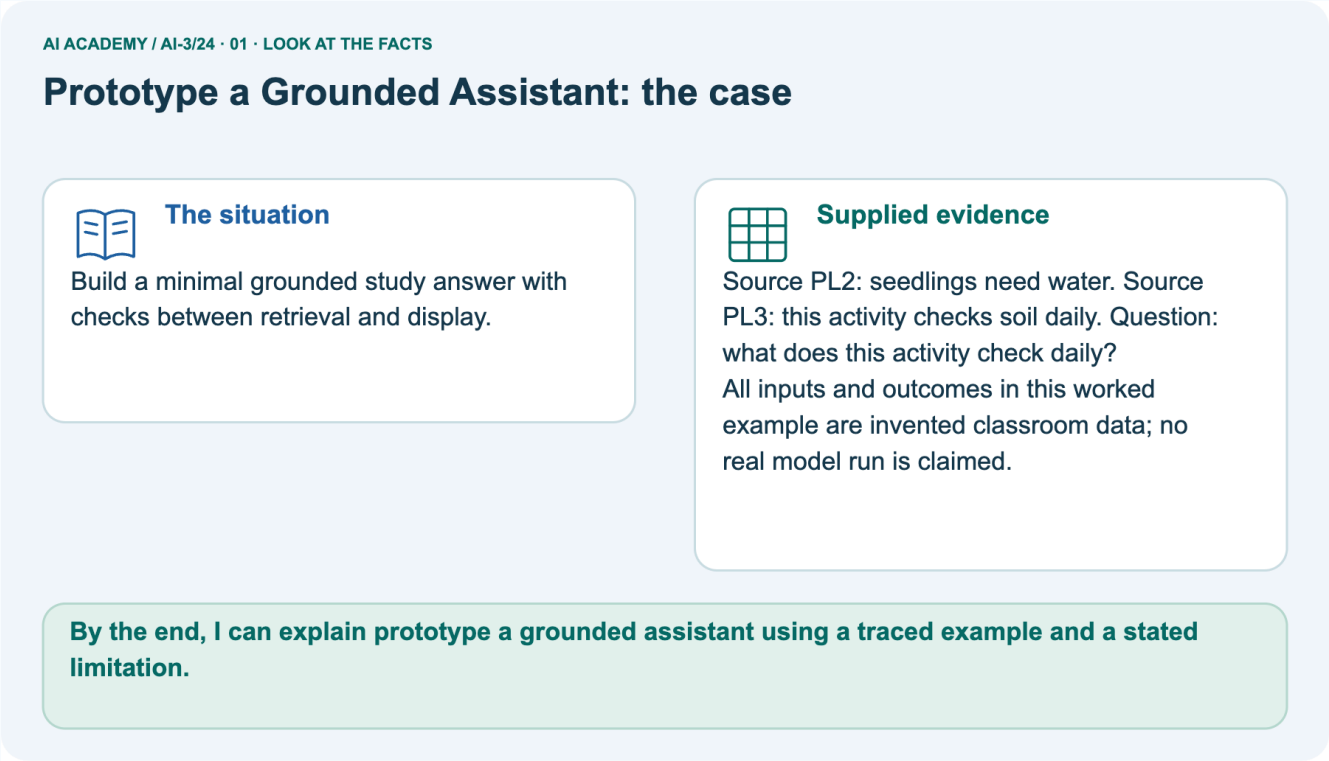
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Prototype a Grounded Assistant: the case



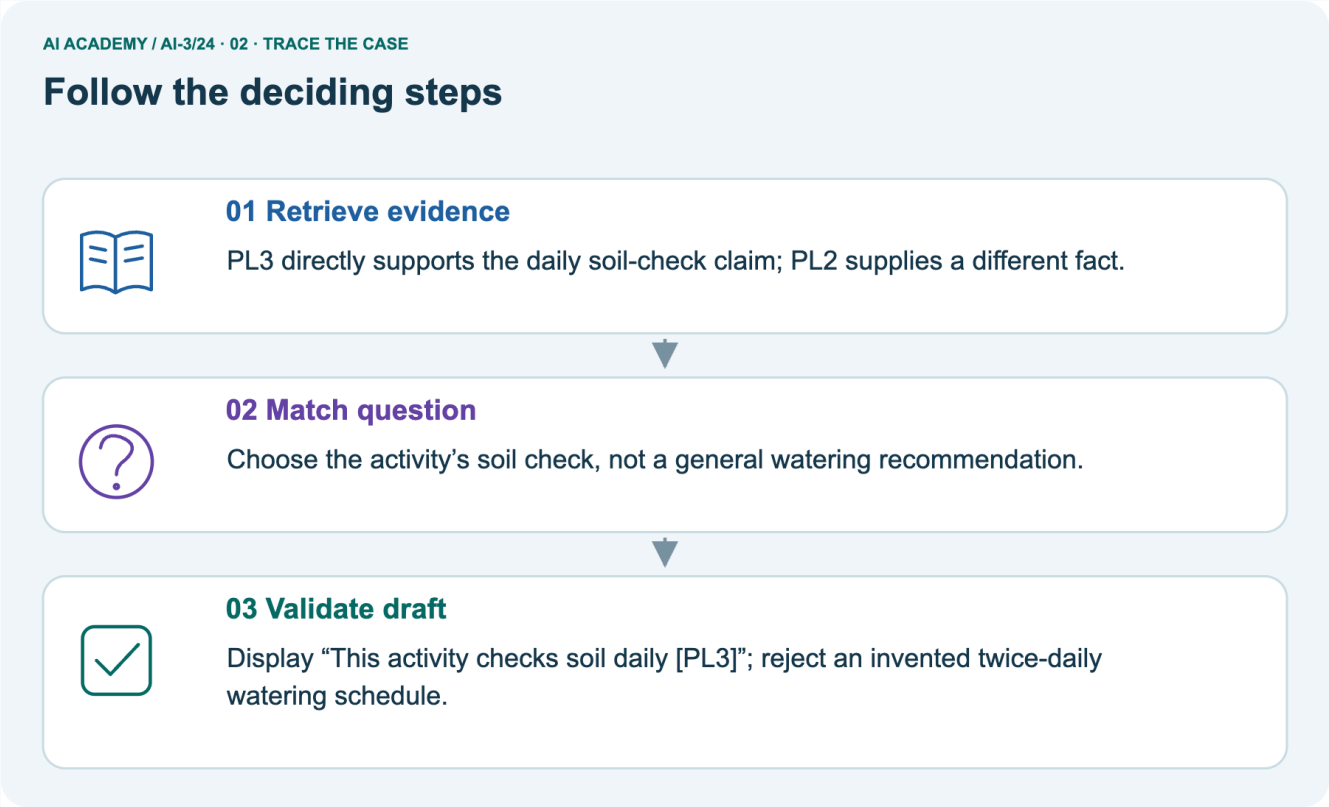
Follow the labelled picture with the matching explanation. Open the vector diagram

Build a minimal grounded study answer with checks between retrieval and display. Supplied facts: Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

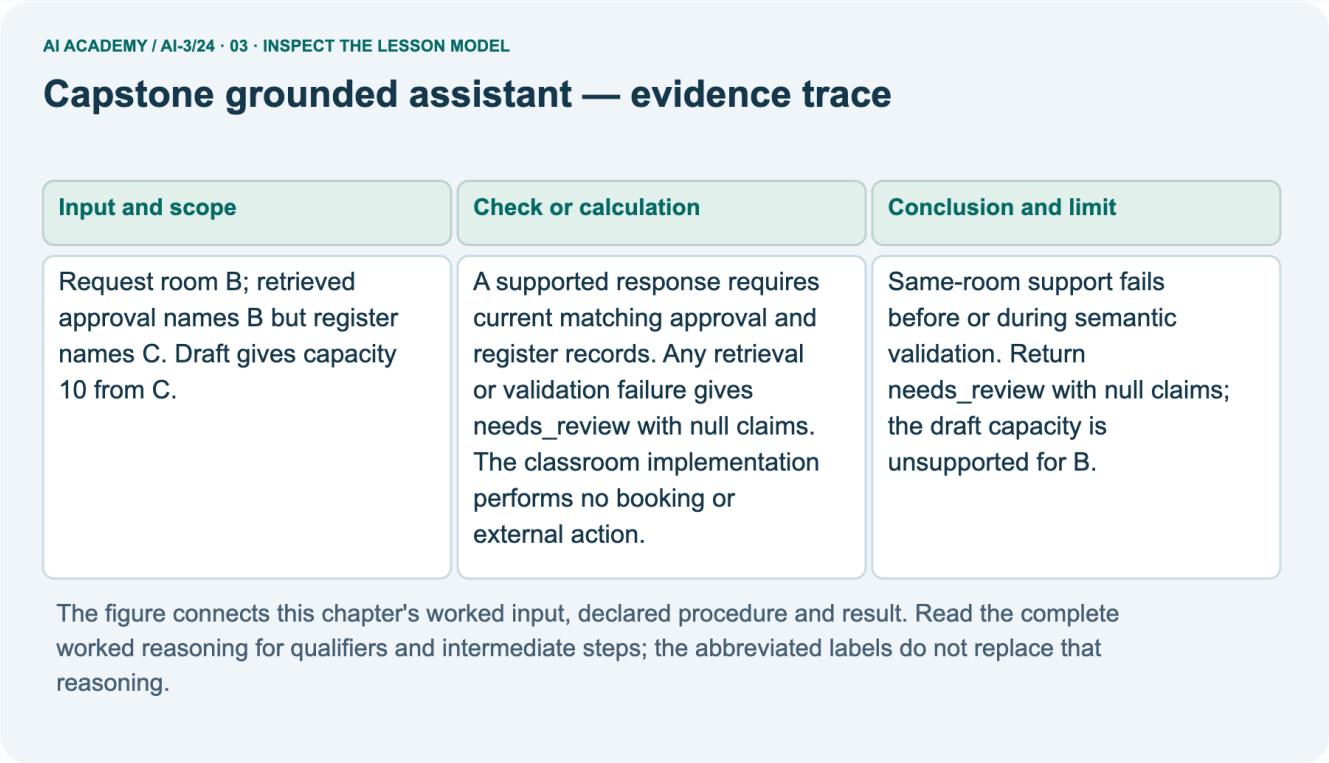


Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve **evidence**: PL3 directly supports the daily soil-check claim; PL2 supplies a different fact. → Match question: Choose the activity's soil check, not a general watering recommendation. → Validate draft: Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone grounded assistant — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

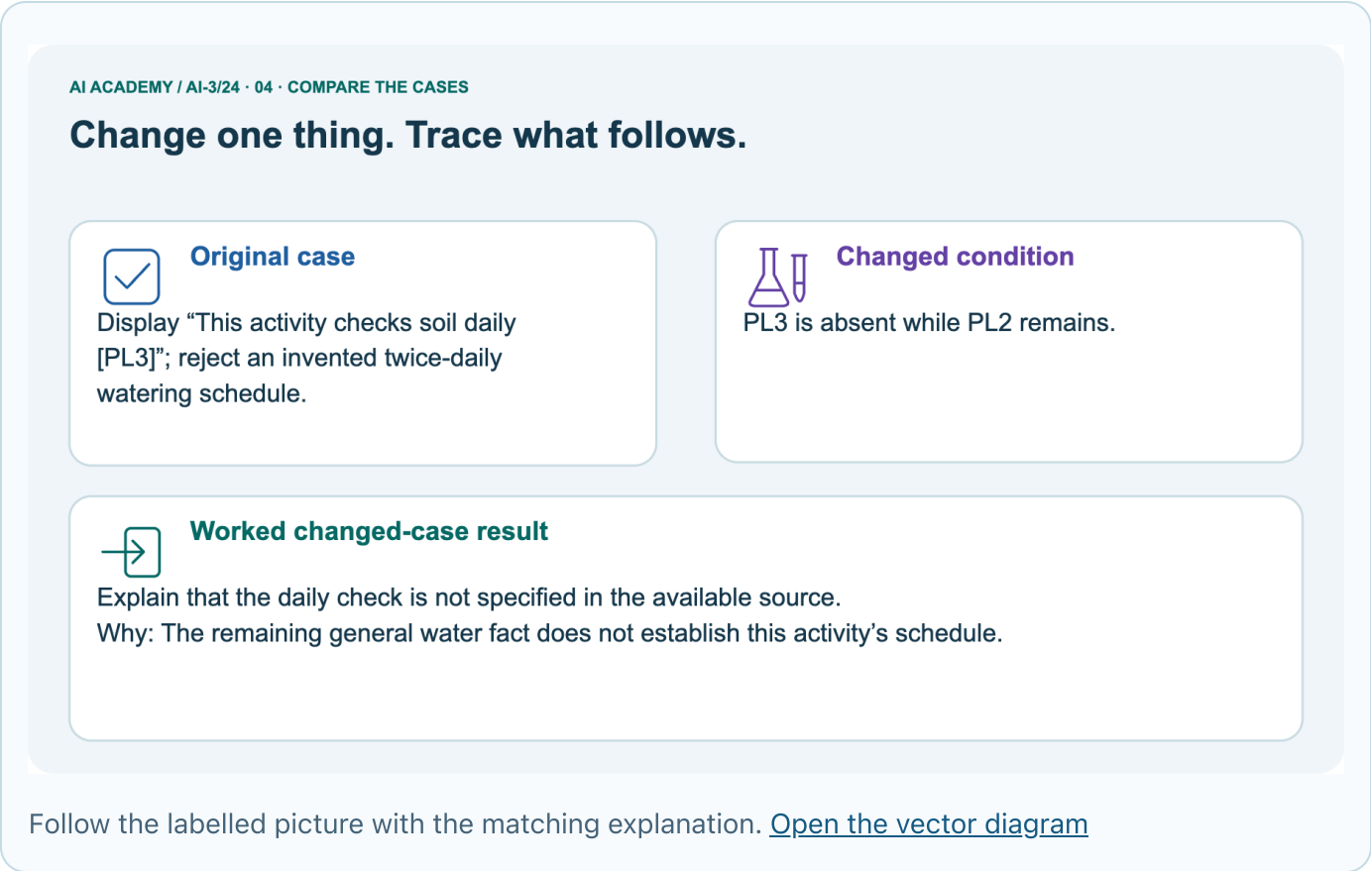
Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



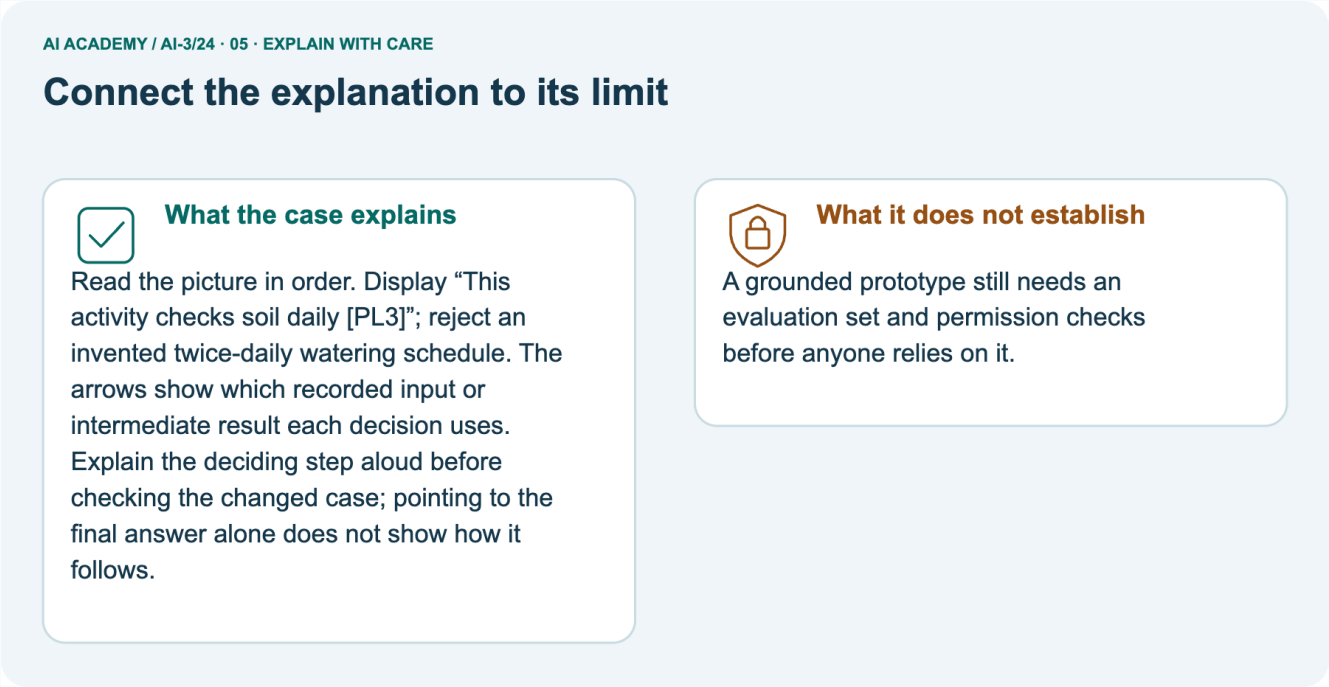
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Changed condition: PL3 is absent while PL2 remains.

Expected result: Explain that the daily check is not specified in the available source. Reason: The remaining general water fact does not establish this activity’s schedule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

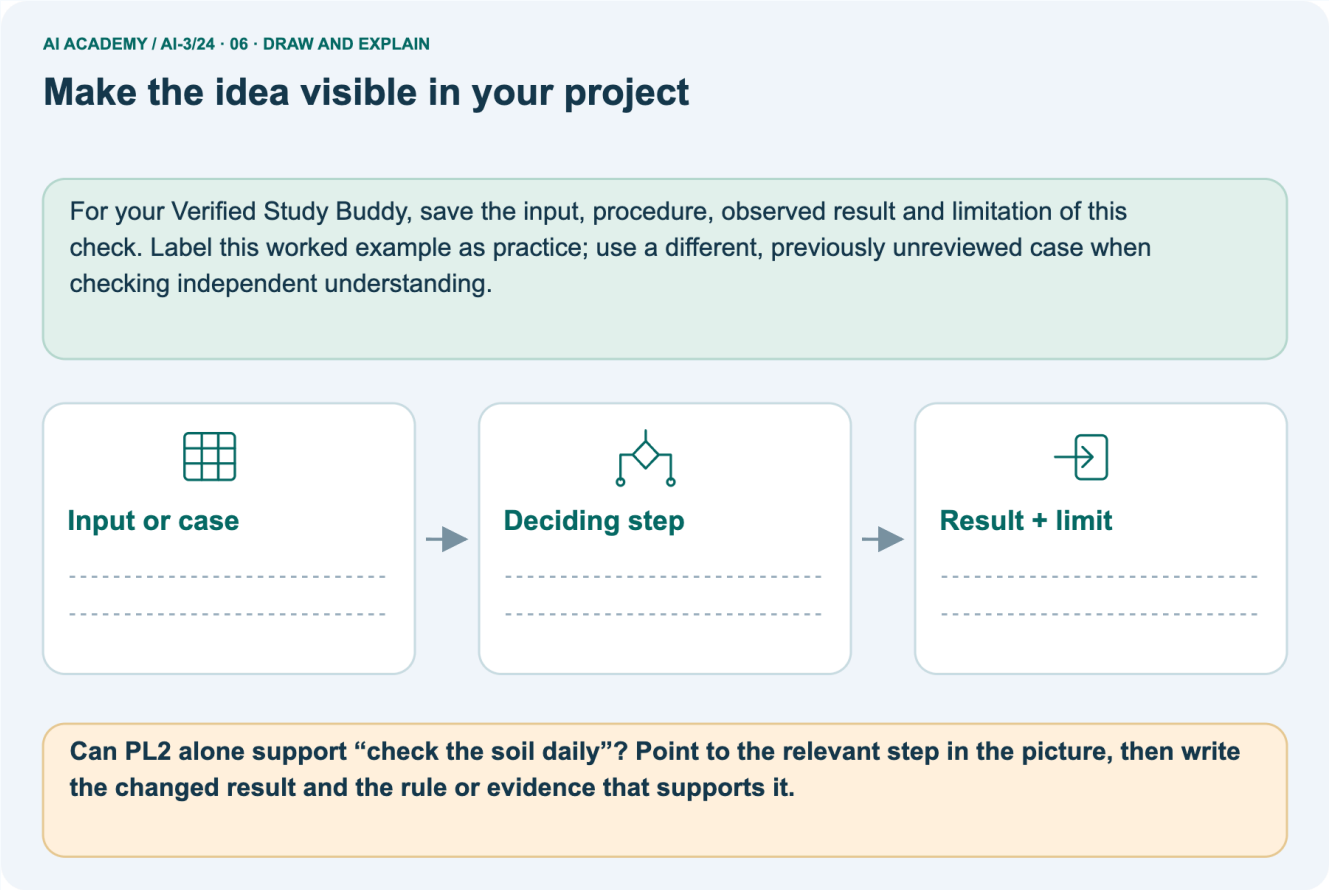
Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Prototype a Grounded Assistant: the case

02. Follow the deciding steps

03. Capstone grounded assistant — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.